Overview of the Internet Architecture & Performance Metrics

Coverage: Network Edge, Network Core, Packet Switching vs Circuit Switching, Delay Models, Loss & Bottleneck Throughput

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1. Internet Structural Architecture

- Network Edge: End systems/hosts (clients, servers, IoT) running application-layer software. Access networks (DSL, FTTH, 5G, Wi-Fi) connect edge hosts to the core mesh.
- Network Core: Interconnected mesh of packet switches (routers and switches) forwarding packets using routing protocols across Tier-1 ISPs, IXPs, and Regional ISPs.
- Standards: Governed by IETF RFCs and IEEE 802 LAN/MAN standards.

2. Packet Switching vs Circuit Switching

- Packet Switching: Data divided into discrete packets. Uses statistical multiplexing on demand. High efficiency for bursty data, but incurs queuing delays.
- Circuit Switching: Dedicated end-to-end physical channel pre-allocated via FDM or TDM. Zero queuing delay, but bandwidth is wasted during idle periods.

3. Mathematical Delay Components

- Total Nodal Delay: d_nodal = d_proc + d_queue + d_trans + d_prop
- Processing Delay (d_proc): Microseconds to inspect header and verify checksums.
- Queuing Delay (d_queue): Waits in output queue. Traffic intensity I = (L * a) / R. As I -> 1, queue delay -> infinity.
- Transmission Delay (d_trans): d_trans = L / R seconds (L = packet bits, R = link rate bps).
- Propagation Delay (d_prop): d_prop = d / s seconds (d = distance, s = propagation speed ~2*10^8 m/s).

■ Worked Problem: Nodal Delay Calculation

- Given: L = 1500 bytes = 12,000 bits. R = 10 Mbps = 10^7 bps. d = 100 km. s = 2 * 10^8 m/s. d_proc = 5 microseconds. 2 packets in queue.
- 1. d_trans = 12,000 / 10^7 = 1.2 ms | 2. d_prop = 10^5 / (2 * 10^8) = 0.5 ms | 3. d_queue = 2 * 1.2 = 2.4 ms.
- 4. Total d_nodal = 0.005 + 2.4 + 1.2 + 0.5 = 4.105 ms.

■ KTU University High-Yield Exam Questions

- Q: Differentiate between packet switching and circuit switching with respect to resource allocation and delay. (7 Marks)
- Q: Derive the four components of nodal delay with mathematical formulas. (7 Marks)